

# CDK2-CCNE MM-GBSA Pipeline

## Session Chat Transcript

Session ID: max-5d0f37323b

Phase: Fix 11 -- TER at ASN245/PRO253 gap + LUMI allocation diagnosis

Cluster: LUMI (lumi.csc.fi) | project\_462001483 | amber/24-cpu

*Generated: 2026-09-29 07:15 UTC*

Model: claude-sonnet-4-6 | Run: max-5d0f37323b | Task: pipeline test job 15

# 1. Session Overview

This document is a complete transcript of the Rayca platform session max-5d0f37323b, which is a continuation of session 90fb861f-3e28-4106-a16d-37c1201f044c after context compaction. The session focuses on debugging and submitting a single-trajectory MM-GBSA binding free energy calculation pipeline for 94 CDK2-CCNE DRAGONFLY de novo ligands on the LUMI HPC cluster (CSC, Finland), using Amber24 CPU (pmemd.MPI, sander) with GAFF2/ff14SB force fields and OBC2 implicit solvent (igb=5, saltcon=0.15 M).

The session contains:

- Pre-compaction history: Fixes 1-10 across 12 pipeline test jobs (summarised)
- Post-compaction turns: Fix 11 diagnosis, LUMI submission debugging, allocation discovery
- Extended thinking: Claude's internal reasoning at each diagnostic step
- Tool calls and cluster query results

| Parameter        | Value                                                                |
|------------------|----------------------------------------------------------------------|
| Session / Run ID | max-5d0f37323b                                                       |
| Previous session | 90fb861f-3e28-4106-a16d-37c1201f044c (compacted)                     |
| Model            | claude-sonnet-4-6                                                    |
| Task             | Fix 11: TER at ASN245/PRO253 gap; pass PIPELINE_TEST_PASSED          |
| Cluster          | LUMI (lumi.csc.fi), project_462001483                                |
| Software         | amber/24-cpu (Amber24, recompiled 2026-02-03 by CSC)                 |
| Method           | Single-trajectory MM-GBSA, GAFF2/ff14SB, TIP3P 12 A, igb=5           |
| Date             | 2026-09-29                                                           |
| Outcome          | BLOCKED -- CPU allocation exhausted (200.1%, 20006/10000 core-hours) |

## 2. Pre-Compaction Session History (Summarised)

### 2.1 Primary Objective

Debug and pass the single-trajectory MM-GBSA pipeline test for mol001 (one of 94 CDK2-CCNE de novo DRAGONFLY ligands). Each test job must exit with PIPELINE\_TEST\_PASSED before a 94-compound production job (cpus=128, gpus=0, minutes=4320) can be submitted. The pipeline prepares the CDK2-CCNE protein-ligand complex, runs 20 ps of MD (minimize+heat+NPT+NVT), strips solvent, computes sander GB single-point energies, and calculates:

$$\Delta G_{\text{MMGBSA}} = \langle E_{\text{complex}} - E_{\text{receptor}} - E_{\text{ligand}} \rangle$$

using VDWAALS + EELEC + EGB + ESURF energy terms.

### 2.2 Infrastructure Fixes (Fixes 1-10)

- Fix 1**

LUMI GPU GRES incompatible -- SLURM registers GPUs as gpu:mi250:8(S:0-7) but platform generates --gres=gpu:N. Resolution: gpus=0, use pmemd.MPI (CPU) on small partition.
- Fix 2**

Input staging path -- platform stages at \$PWD not \$RAYCA\_OUT. Resolution: STARTDIR=\$PWD before cd; tar from \$STARTDIR.
- Fix 3**

antechamber odd-electron charge -- mol001 nominal charge +1 gives 139 valence electrons (odd), sqm Fatal Error. Resolution: retry charge-1 then charge+1; charge 0 (140 electrons) succeeds.
- Fix 4**

Platform set -e suppresses retry logic -- wrapper prepends set -euo pipefail; semicolon pattern fires errexit. Resolution: set +e after module loads; use || pattern.
- Fix 5**

tleap ion neutralization -- additions Na+ 0 then Cl- 0 fails same-sign; additions2 with 0+ion2 invalid. Resolution: compute SYS\_CHARGE at runtime, write single explicit-count additions2 line.
- Fix 6**

saveprmtop / saveambercrd invalid in Amber24 tleap. Resolution: saveAmberParm COMPLEX complex.prmtop complex.inpcrd.
- Fix 7/8**

pdb4amber absent on LUMI amber/24-cpu. Resolution: inline Python heredoc for receptor preprocessing (4 passes).
- Fix 9**

Spurious OXT on non-C-terminal ARG B359 (H++ artifact). Resolution: Pass 0 detects true C-termini per chain; strips OXT from non-C-terminal residues.
- Fix 10**

ALA B360 bad geometry -- H++ placed OXT 3.26 Å from C and N 3.01 Å from ARG B359C, creating duplicate 1-4 exclusion path in tleap. Resolution: SKIP\_RESIDUES={{B,360}}; ARG B359 becomes C-terminal.

### 2.3 Pipeline Test Job History

| Job ID   | CPUs | GPU | Wall   | Failed at     | Exit cause                                             |
|----------|------|-----|--------|---------------|--------------------------------------------------------|
| 21925395 | 64   | 8   | 30 min | scheduler     | GPU GRES incompatible                                  |
| 21925247 | 16   | 0   | 10 min | tar           | Input file at \$PWD not \$RAYCA_OUT                    |
| 21925962 | 16   | 0   | 45 min | antechamber   | Odd-electron charge=1                                  |
| 21926492 | 16   | 0   | 45 min | antechamber   | Platform set -e fires before retry                     |
| 21926562 | 16   | 0   | 45 min | tleap         | additions same-sign conflict                           |
| 21926682 | 16   | 0   | 45 min | tleap         | additions2 with 0 + ion2 invalid                       |
| 21926974 | 16   | 0   | 45 min | tleap         | saveprmtop invalid (Amber24)                           |
| 21930902 | 16   | 0   | 45 min | tleap         | 20 FATAL atom-type errors (H++ H atoms + HIS names)    |
| 21931141 | 16   | 0   | 45 min | receptor prep | pdb4amber not installed (exit 127)                     |
| 21933965 | 16   | 0   | 45 min | tleap         | ARG B359 OXT has no type (H++ artifact)                |
| 21935712 | 16   | 0   | 45 min | tleap         | FATAL 1-4 bond 9264/9265 -- ALA B360 bad geometry      |
| 21941002 | 16   | 0   | 45 min | tleap         | FATAL 1-4 bond 9254/9255 -- ASN457/PRO458 gap (15.4 Å) |
| job15    | 16   | 0   | 45 min | --            | BLOCKED: CPU allocation exhausted (200.1%)             |

### 3. Pre-Compaction Conversation Turns (Reconstructed from Summary)

USER

what is the update

CLAUDE (internal thinking)

Session restart after context compaction. User wants current status. I need to read the run manifest and last documents to understand where we left off.

The manifest says: Run max-5d0f37323b, Task: Fix 11 (TER at ASN245/PRO253 gap), Phase: pass pipeline test (id 1), Open obligations: recovery\_audit.

Job 21941002 (Fix 10) was submitted. I need to check its result.

CLAUDE

Checking the run manifest and last job result. The manifest shows Fix 11 is the active task -- inserting a TER between ASN245 and PRO253 in chain B to resolve the 15.4 Å loop gap that tleap builds as a covalent bond, causing the 1-4 FATAL at bond 9254/9255.

Job 21941002 (Fix 10: drop ALA B360) FAILED. The 1-4 FATAL reappeared at bond 9254/9255 (shifted -10 from 9264/9265 after removing 10 atoms in ALA B360). This confirms the ASN245/PRO253 loop gap is the true remaining cause.

Fix 11 (Pass 3) adds TER between non-consecutive residue numbers within the same chain.  $CASN(-1) + NPRO(+1) = 0$  net charge; RECEPTOR\_CHARGE stays -4. Script updated, echo: 'job15 Fix11'. Ready to submit.

USER

update

CLAUDE

Fix 11 code is written and incorporated into pipeline\_test\_mol001.sh. Attempted submission to LUMI failed with AssocMaxSubmitJobLimit. Queue confirmed empty (squeue returns 0 jobs). Investigating why SLURM refuses submissions despite empty queue. Will retry -- this may be a transient accounting lag after job 21941002 ended.

USER

CRITICAL: Respond with TEXT ONLY. [Session summary request -- pre-compaction]

CLAUDE

[Session summary provided to compaction system. Content covers: Fixes 1-11, job history through 21941002, Fix 11 code details, AssocMaxSubmitJobLimit blocker, empty queue confirmation, ready-to-submit state.]

#### 4. Post-Compaction Conversation (Full Transcript)

Turn 1

USER

results?

CLAUDE (internal thinking)

Session has been restarted after context compaction. The system reminder says:

Run: max-5d0f37323b (running)

Task: Fix 11 -- insert TER at ASN245/PRO253 gap, pass pipeline test (id 1)

Files produced: 0

Open obligations: recovery\_audit

The previous attempt to submit job 15 (Fix 11) failed with AssocMaxSubmitJobLimit. Queue was confirmed empty. I need to investigate this further -- why does SLURM refuse even though the queue is empty?

Let me check: squeue state + sacctmgr association limits + recent sacct history.

CLAUDE

Persistent AssocMaxSubmitJobLimit -- querying account details to find the actual constraint before retrying blind.

[Tool: query\_cluster LUMI]

Command: squeue -u \$USER; sacctmgr show assoc where user=\$USER; sacct --starttime=\$(date -d '-1 hour')

Turn 1 -- LUMI squeue result

```
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
[empty -- 0 jobs]

sacctmgr associations:
default_n+      0      0
project_4+      0      2
project_4+      2      2      <-- MaxSubmitJobs=2 on one partition
project_4+      0      1
project_4+     210    200
project_4+      0    200      <-- small partition: MaxSubmit=0 (unlimited)
...

sacct: no jobs in last 1 hour
```

CLAUDE (internal thinking)

Queue is empty. The MaxSubmitJobs=2 on one association is concerning but  $0 < 2$ , so it shouldn't block. The 'default\_no\_jobs' account name is suspicious.

The sacct shows no jobs in the last hour -- but job 21941002 ran a WEEK ago (Sept 11). Need to check sacct over a wider window to see if any ghost jobs are counted.

Also: GrpSubmit is a GROUP limit, not per-user. If other project members have jobs...

Let me check recent jobs from Sept 15+ and the current PENDING/RUNNING/COMPLETING state.

Turn 2 -- Wider sacct window

```
sacct --starttime=2026-09-15:
22074860 standard FAILED 00:00:00 2026-09-15T18:16
22074924 standard FAILED 00:00:00 2026-09-15T18:21
22074947 standard FAILED 00:00:00 2026-09-15T18:22
```

```
22074955 standard COMPLETED 00:00:05 2026-09-15T18:24
22075127 standard TIMEOUT 2-00:00 2026-09-15T18:30 <-- 2-day job
22075127.6 CANCELLED 02:26:37 2026-09-17T16:04
22076857 standard TIMEOUT 2-00:00 2026-09-15T20:22 <-- 2-day job
22076857.21 CANCELLED 01:59:49 2026-09-17T18:22
... many more standard-partition jobs ...

squeue --states=PENDING,RUNNING,COMPLETING: [empty]
```

**CLAUDE (internal thinking)**

Two 2-day standard-partition jobs (22075127, 22076857) both TIMED OUT on Sept 15-17. Last step cancelled 2026-09-17T18:22. We're now Sept 18. These should not block new submissions.

GrpSubmit on standard=100, MaxSubmit=0. With 0 jobs in queue, this can't be it.

The 'default\_no\_jobs' account -- could SLURM be defaulting to that if the platform doesn't specify --account=project\_462001483? Let me check the user's default account.

### Turn 3 -- Partition/QOS details

```
sacctmgr show assoc where user=$USER partition=small:
project_4+  small  normal  0  200
(MaxSubmit=0 = unlimited, GrpSubmit=200)

QOS 'normal': no MaxSubmitJobsPerUser set

sinfo -p small: 103 mixed, 188 allocated nodes -- partition is UP
```

#### CLAUDE (internal thinking)

small partition: MaxSubmit=0 (unlimited per-user), GrpSubmit=200. That's fine.

QOS 'normal' shows no per-user submit limits.

The partition is up and accepting jobs.

Then WHY is it refusing? Let me re-submit and see if AssocMaxSubmitJobLimit persists. Maybe it was a transient rate limit that's cleared by now. Also let me check: what is the user's default account?

### Turn 4 -- Re-submit job 15 (Fix 11)

#### USER

[Platform submits job 15 with all Fixes 1-11]

```
ERROR: cluster_failed
sbatch: error: AssocMaxSubmitJobLimit
sbatch: error: Batch job submission failed: Job violates accounting/QOS policy
(job submit limit, user's size and/or time limits)
retryable: false
```

#### CLAUDE (internal thinking)

Still blocked. Even though the queue is empty. This is not a transient lag.

The platform marks this retryable=false. Something structural is wrong.

New hypothesis: Maybe the platform is submitting to the wrong account/partition, or there's a QOS that restricts this specific job type. Let me try the absolute minimum possible job (1 CPU, 2 minutes, echo only) to rule out job size as a factor.

### Turn 5 -- Trivial probe job

#### CLAUDE

Even a 1-CPU/2-minute echo job is refused. This confirms the issue is not our script -- ALL LUMI submissions from this account are currently blocked.

[Trivial probe: cpus=1, gpus=0, minutes=2, script='echo test']

[Result: AssocMaxSubmitJobLimit -- same error]

#### CLAUDE (internal thinking)

All submissions are refused. Even 1 CPU / 2 minutes. This is global.

Next step: check what account previous successful jobs used, then check the user's default account and group-wide queue.

### Turn 6 -- Check previous job accounts

```
sacct -j 21941002,21935712,21933965 -o jobid,account,partition,state:
21933965  project_4+  standard  FAILED
21935712  project_4+  standard  FAILED
21941002  project_4+  standard  FAILED
```

All previous jobs used account=project\_462001483, partition=standard.

**CLAUDE (internal thinking)**

Previous jobs used project\_462001483 on standard partition. Same as what should happen now. So the account routing is correct.

New focus: is the default account 'default\_no\_jobs' causing issues?

Let me check: sacctmgr show user withassoc to see defaultaccount.

Also: the standard partition GrpSubmit=100 is a GROUP limit. Maybe OTHER users in project\_462001483 have filled the group quota?

Let me check: squeue -A project\_462001483



## 4 (continued) -- Discovery of Root Cause

### Turn 7 -- Full sacctmgr associations + default account

```
sacctmgr show user $USER withassoc format=user,defaultaccount:
matimuha    default_no_jobs    default_no_jobs
matimuha    default_no_jobs    project_462001483
...

DEFAULT ACCOUNT = 'default_no_jobs'

Full association table:
Account      Partition  GrpSubmit  MaxSubmit
default_no_jobs (all)      0          0
project_462001483 debug      2          0
project_462001483 dev-g      2          2 <-- MaxSubmit=2!
project_462001483 interactive 1          0
project_462001483 small-g    200        210
project_462001483 small      200        0 <-- unlimited!
project_462001483 standard-g 200        210
project_462001483 standard  100        0
```

#### CLAUDE (internal thinking)

The default account is 'default\_no\_jobs'. In SLURM, if you don't specify --account, SLURM uses the default account. 'default\_no\_jobs' might block all submissions.

BUT -- previous jobs all ran on project\_462001483. So the platform was correctly specifying --account=project\_462001483. And if it's still doing that, the default account shouldn't matter.

The only partition with MaxSubmit=2 is dev-g. We're not submitting there.  
small partition: MaxSubmit=0 (unlimited), GrpSubmit=200.

I'm stuck. The limits all look fine, the queue is empty.

NEW IDEA: Check group-wide queue. GrpSubmit=100 on standard is for ALL project members. Maybe other users in the project have filled it? Let me check.

### Turn 8 -- Group-wide queue check

```
squeue -A project_462001483 --all:
JOBID PARTITION USER ST TIME NODES
[empty]

Total jobs for project_462001483: 0
```

#### CLAUDE (internal thinking)

0 jobs for the ENTIRE project. Not a group quota issue.

I'm running out of SLURM-level explanations. With:

- 0 user jobs
- 0 project-wide jobs
- MaxSubmit=0 (unlimited) on small/standard partitions
- GrpSubmit well under limits

The only thing I haven't checked: LUMI's billing allocation system. LUMI uses CSC billing units (BUs). When a project exhausts its compute allocation, SLURM refuses all submissions with AssocMaxSubmitJobLimit.

The two recent 2-day jobs (22075127, 22076857) each ran for 2 days on standard partition. That's approximately 2 \* 2 \* 128 CPUs \* 24h = thousands of core-hours. This could have exhausted the allocation.

Let me check: lumi-allocations

Turn 9 -- Allocation check (ROOT CAUSE FOUND)

```
lumi-allocations:
Data updated: 2026-09-18 11:37:41

Project          | CPU (used/allocated)          | GPU (used/alloc)          | Storage
project_462001483 | 20006/10000 (200.1%) core/hours | 12/1000 (1.2%) GPU-h     | 248/10000 (2.5%) TB-h

lumi-quota:
/scratch/project_462001483  114G/55T  228K/2.0M files
/projappl/project_462001483  8.2K/54G  2/100K files
/users/matimuha              17G/22G   68K/100K files
```

CLAUDE (internal thinking)

FOUND IT.

CPU allocation: 20,006 / 10,000 core-hours = 200.1% OVER BUDGET.

The two recent 2-day standard-partition jobs (22075127, 22076857) burned through the remaining allocation and pushed it 100% over. LUMI enforces a hard cutoff: once the allocation is exceeded, SLURM refuses ALL new submissions with AssocMaxSubmitJobLimit, regardless of queue state, job size, or partition.

This explains everything:

- Empty queue: existing jobs already finished (TIMEOUT/CANCELLED)
- All submissions refused: allocation enforcement at account level
- Even 1-CPU/2-minute jobs refused: the check happens at submission, not at dispatch
- GPU allocation fine (1.2%): only CPU was burned

This is not fixable by changing script parameters. The project needs a CPU allocation top-up from CSC via the MyCSC portal.

**Turn 9 -- Claude final response****CLAUDE**

Found the blocker. The project CPU allocation is EXHAUSTED at 200.1% -- 20,006 core-hours used against a 10,000 core-hour budget. SLURM refuses all submissions when the allocation is over limit, which is why even a 1-CPU/2-minute probe fails.

ROOT CAUSE: Two recent standard-partition 2-day jobs (22075127, 22076857) burned the remaining allocation and pushed it 100% over. LUMI enforces a hard cutoff: once exceeded, AssocMaxSubmitJobLimit fires on every submit regardless of queue state.

WHAT'S NEEDED TO UNBLOCK:

1. Request a CPU allocation top-up for project\_462001483 from CSC:
  - Via MyCSC portal: <https://my.csc.fi> (resource management page)
  - Or email: servicedesk@csc.fi citing project number + 200% over
2. GPU allocation still has headroom (1.2% used, 12/1000 gpu-hours) but Amber CPU jobs cannot run on GPU nodes.

ONCE ALLOCATION IS RESTORED:

Job 15 (Fix 11) is ready to go -- no script changes needed.

Submit: cpus=16, gpus=0, minutes=45

Script: pipeline\_test\_mol001.sh (all Fixes 1-11)

Expected: 4 TER records, tleap zero FATALs, PIPELINE\_TEST\_PASSED

**Task #1 -- Updated status**

```
Task #1: Fix 11 -- insert TER at ASN245/PRO253 gap, pass pipeline test
Status: in_progress (BLOCKED on LUMI allocation)
Script: pipeline_test_mol001.sh -- READY, no changes needed
Blocker: project_462001483 CPU at 200.1% (20006/10000 core-hours)
Action required: CSC allocation top-up via MyCSC portal
```

**5. Fix 11 Technical Detail**

Fix 11 resolves the final tleap 1-4 FATAL error by inserting a TER record between ASN B245 and PRO B253 in the receptor PDB before passing it to tleap.

ROOT CAUSE: The CDK2-CCNE crystal structure has a disordered loop in chain B (CCNE, residues 246-252 absent). H++ produced no TER between ASN245 and PRO253, leaving a 15.392 Å gap that tleap treats as a covalent peptide bond. During saveAmberParm topology build, the long-range bond creates invalid 1-4 exclusion pairs (bond 9254/9255 in job 21941002, after -10 from removing ALA B360's 10 atoms).

SOLUTION (Pass 3 in the Python heredoc):

After Pass 2 builds the clean atom list, scan for ATOM records where the residue number jumps by >1 within the same chain. Insert 'TER\n' before the first atom of each gap.

CHARGE IMPACT:

Inserting TER creates: CASN (C-terminal asparagine, COO- terminus, -1) and NPRO (N-terminal proline, NH3+ terminus, +1). Net charge change = 0. RECEPTOR\_CHARGE remains -4. Ion count unchanged: additions2 COMPLEX Na+ 4.

EXPECTED RECEPTOR OUTPUT:

4585 heavy atoms (unchanged from job 21941002)

4 TER records (was 3 -- new TER at chain B gap 245->253)

C-term set: [('A',298), ('B',359)]

OXT atoms kept: [('A',298,'LEU'), ('B',359,'ARG')]

HID (15), HIE (3) [unique residues, not atom count]

TER inserted: chain B gap 245->253

### Pass 3 Python code (core logic)

```
# Pass 3: TER at internal chain breaks (ASN245/PRO253 gap, 15.4 A)
final = []
prev_chain = None
prev_resnum = None
for l in out:
    if l[:4] == 'ATOM':
        chain = l[21]
        try: resnum = int(l[22:26])
        except ValueError:
            final.append(l); continue
        if prev_chain == chain and prev_resnum is not None and resnum - prev_resnum > 1:
            final.append('TER\n')
            print(f'  TER inserted: chain {chain} gap {prev_resnum}->{resnum}', flush=True)
            prev_chain = chain; prev_resnum = resnum
            final.append(l)
    else:
        final.append(l)
        if l[:3] == 'TER': prev_chain = None; prev_resnum = None
out = final
```

## 6. Report Updated This Session

The phase report was updated at the end of this session:

File: reports/phase\_03\_submit\_mmgbsa\_94\_cdk2\_ccne\_ligands\_lumi.md

Changes made:

- Status line: updated to BLOCKED with allocation percentage
- Fix 10 entry: added actual job 21941002 result (FAILED, bond 9254/9255 shifted by -10)
- Fix 11 entry: full description of Pass 3, charge impact, expected output
- Pipeline test table: job 21941002 marked FAILED; job 15 marked BLOCKED
- New blocker section: LUMI CPU allocation at 200.1%, resolution steps
- Next step: updated to include allocation top-up before job 15 submission
- Verification table: tleap/pmemd/cpptraj/sander rows updated to 'pending'

## 7. Current State Summary

| Parameter                | Value                                                               |
|--------------------------|---------------------------------------------------------------------|
| Pipeline test script     | pipeline_test_mol001.sh -- Fixes 1-11 ALL incorporated              |
| Fix 11                   | Pass 3 TER insertion at chain B gap 245->253 -- READY               |
| LUMI CPU allocation      | 20,006 / 10,000 core-hours = 200.1% EXHAUSTED                       |
| LUMI GPU allocation      | 12 / 1,000 gpu-hours = 1.2% (plenty available)                      |
| Job 15 submission status | BLOCKED -- all LUMI submissions refused                             |
| Error on submit          | AssocMaxSubmitJobLimit (allocation enforcement)                     |
| Queue state              | 0 jobs for user matimuha, 0 jobs for project_462001483              |
| Action required          | CPU allocation top-up via MyCSC portal or CSC servicedesk           |
| After top-up             | Submit job 15: cpus=16, gpus=0, minutes=45, no script changes       |
| After test passes        | Update mmgbsa_production.sh prep_compound(), submit 94-compound job |
| Production job params    | cpus=128, gpus=0, minutes=4320 (3 days), small partition            |
| Production expected      | 8 parallel compounds x 16 CPUs; 24-48 h wall-clock                  |